

Greenhouse Gas Emissions Inventory Report

Prepared for

Northwind Manufacturing, Inc.

Reporting period	1 January - 31 December 2025
Reporting standard	GHG Protocol Corporate Accounting & Reporting Standard
Regulatory basis	California SB 253 - Climate Corporate Data Accountability Act
Report generated	7 July 2026, 03:18 UTC

Total reported emissions

136.59 tCO2e

This report presents a greenhouse gas emissions inventory prepared using the GHG Protocol Corporate Standard. Figures are self-reported based on activity data supplied by the organization and are structured to support third-party limited assurance as required under California SB 253. This document is not itself an assurance opinion.

1. Reporting Entity & Organizational Boundaries

Reporting entity	Northwind Manufacturing, Inc.
Reporting period	1 January - 31 December 2025
Consolidation approach	Operational control
Reporting standard	GHG Protocol Corporate Accounting & Reporting Standard (revised)
Gases covered	CO ₂ , CH ₄ , N ₂ O, HFCs, PFCs, SF ₆ , NF ₃ (7 Kyoto gases), expressed as CO ₂ e

The organizational boundary is defined using the operational control approach: emissions from operations over which the entity has the authority to introduce and implement its operating policies are consolidated at 100%. Emissions are classified into Scope 1 (direct), Scope 2 (purchased energy) and Scope 3 (value chain) in accordance with the GHG Protocol.

2. Executive Summary

For the 2025 reporting year, Northwind Manufacturing, Inc. and its consolidated operations reported total greenhouse gas emissions of 136.59 tonnes of CO₂ equivalent (tCO₂e), disaggregated by scope below.

Scope	Emissions (tCO ₂ e)	Share
Scope 1 - Direct	32.95	24.1%
Scope 2 - Energy indirect	73.75	54.0%
Scope 3 - Value chain	29.88	21.9%
Total	136.59	100%

3. Scope 1 Emissions by Greenhouse Gas

The GHG Protocol requires the seven Kyoto gases to be reported. Scope 1 combustion emissions are disaggregated into CO₂, CH₄ and N₂O below, expressed as CO₂e using IPCC AR6 100-year GWPs (CH₄ fossil = 29.8, N₂O = 273). The industrial gases (HFCs, PFCs, SF₆, NF₃) arise from fugitive/process sources; none were reported in this inventory.

Greenhouse gas	Emissions (tCO ₂ e)	Share of Scope 1
Carbon dioxide (CO ₂)	32.54	98.8%
Methane (CH ₄)	0.10	0.3%
Nitrous oxide (N ₂ O)	0.31	0.9%
Hydrofluorocarbons (HFCs)	0.00	0.0%
Perfluorocarbons (PFCs)	0.00	0.0%
Sulfur hexafluoride (SF ₆)	0.00	0.0%
Nitrogen trifluoride (NF ₃)	0.00	0.0%
Total Scope 1 (fossil)	32.95	100%

The combustion split uses representative fossil-fuel gas shares (CO₂-dominant); for third-party assurance, apply fuel- and technology-specific gas factors (EPA GHG Emission Factors Hub / IPCC). Biogenic CO₂ of 2.14 tCO₂e (from biofuels) is reported separately and excluded from the Scope 1 fossil total, per the GHG Protocol.

4. Scope 2 Dual Reporting (Location & Market-Based)

The GHG Protocol Scope 2 Guidance requires purchased-electricity emissions to be reported under both methods: the location-based method (grid-average emission factors) and the market-based method (contractual instruments such as supplier-specific factors, renewable energy certificates, guarantees of origin, or power purchase agreements). Both are presented below.

Scope 2 method	Emissions (tCO ₂ e)
Location-based (grid average)	73.75
Market-based (contractual)	73.75

No contractual instruments (supplier-specific emission factors or renewable/REC coverage) were supplied for the reporting period; the market-based figure therefore equals the location-based figure. To reflect green procurement, provide a per-account supplier emission factor (kgCO₂e/kWh) or a renewable share with your electricity data.

5. Methodology

Emissions were quantified using the activity-based method defined by the GHG Protocol Corporate Standard: each activity's emissions equal its activity data multiplied by a published emission factor for the relevant scope, region and category.

Calculation formulae applied:

- General: Emissions (tCO₂e) = Sum(Activity data x Emission factor x GWP).
- Scope 1 (combustion): Sum(Fuel consumed x Fuel emission factor). Fuel-specific factors bundle CO₂, CH₄ and N₂O expressed as CO₂e.
- Scope 2 (location-based): Sum(Electricity (kWh) x Grid-average emission factor), using US EPA eGRID grid factors.
- Scope 3 travel: Sum(Distance x Mode-specific factor). Scope 3 spend: Sum(Spend x EEIO factor).

Global Warming Potentials	IPCC Sixth Assessment Report (AR6), 100-year time horizon (GWP-100)
Scope 2 methods	Dual reporting: location-based and market-based (see Scope 2 Dual Reporting section)

Biofuels (e.g. biodiesel, ethanol, sustainably-sourced wood) are treated as carbon-neutral at the point of combustion per the GHG Protocol; biogenic CO₂ is excluded from Scope 1 totals and reported separately where present. Emission factors are representative published values from the sources cited; for third-party assurance, confirm the specific factor vintage applicable to the reporting period.

Emission factor sources used in this inventory: US EPA GHG Emission Factors Hub; US EPA eGRID (location-based); UK DEFRA / GHG Protocol (passenger transport); US EPA EEIO (spend-based); UK DEFRA (waste treatment); UK DEFRA / GHG Protocol (freight). A complete factor-by-factor audit trail is provided in a later section.

6. Emissions by Source Category

Category	Scope	Emissions (tCO ₂ e)	Share
Electricity	2	73.75	54.0%
Diesel	1	24.66	18.1%
Car	3	12.66	9.3%

Natural Gas	1	8.30	6.1%
Cloud Services	3	7.04	5.2%
Flight Long Haul	3	6.59	4.8%
Landfill	3	2.52	1.8%
Truck	3	1.07	0.8%

7. Emissions by Location

Location	Emissions (tCO2e)	Share
US-CA	130.00	95.2%
GLOBAL	6.59	4.8%

8. Scope 3 Category Coverage

The GHG Protocol defines 15 Scope 3 categories. Each is assessed below; categories not reported are those for which activity data was not provided for this reporting period and are candidates for future inclusion.

#	Category	Status	tCO2e
1	Purchased goods and services	Reported	7.04
2	Capital goods	Not reported	-
3	Fuel- and energy-related activities	Not reported	-
4	Upstream transportation and distribution	Reported	1.07
5	Waste generated in operations	Reported	2.52
6	Business travel	Reported	6.59
7	Employee commuting	Reported	12.66
8	Upstream leased assets	Not reported	-
9	Downstream transportation and distribution	Not reported	-
10	Processing of sold products	Not reported	-
11	Use of sold products	Not reported	-
12	End-of-life treatment of sold products	Not reported	-
13	Downstream leased assets	Not reported	-
14	Franchises	Not reported	-
15	Investments	Not reported	-

Exclusion basis: categories marked 'Not reported' are excluded due to unavailable primary activity data, not immateriality. Their relevance should be reassessed as data collection matures, consistent with GHG Protocol completeness.

9. Base Year & Emissions Intensity

Base year	2025
Base year rationale	First greenhouse gas inventory; established as the base year for future tracking
Base year total	136.59 tCO2e

Under the GHG Protocol, a base year is fixed as the reference against which future emissions are compared. It will be recalculated for significant structural changes (acquisitions, divestitures, or methodology changes) exceeding the entity's significance threshold.

Emissions intensity normalizes total emissions against a measure of business activity, enabling comparison across years and peers independent of absolute growth.

Emissions intensity metric		Value (tCO2e)
Annual revenue	\$1,200,000,000	
Per \$1M revenue		0.11
Full-time-equivalent headcount	850	
Per full-time-equivalent employee		0.16
Intensity is computed on the location-based total for the reporting period. Under California SB 253, revenue is also the threshold on which reporting applicability is assessed.		

10. Emission Factors & Audit Trail

Every emission factor applied in this inventory is listed below with its published source, supporting third-party verification.

Sc.	Category	Factor	Unit	Source
1	Diesel	2.68	kg/L	US EPA GHG Emission Factors Hub
1	Natural Gas	1.93	kg/therm	US EPA GHG Emission Factors Hub
2	Electricity	0.298	kg/kWh	US EPA eGRID (location-based)
3	Car	0.171	kg/mile	UK DEFRA / GHG Protocol (passenger transport)
3	Cloud Services	0.08	kg/USD	US EPA EEIO (spend-based)
3	Flight Long Haul	0.195	kg/mile	UK DEFRA / GHG Protocol (passenger transport)
3	Landfill	0.467	kg/kg	UK DEFRA (waste treatment)
3	Truck	0.089	kg/tonne-km	UK DEFRA / GHG Protocol (freight)

11. Activity Data

The underlying activity records used to calculate this inventory. Each row is one measured activity and its resulting emissions.

Sc.	Category	Location	Quantity	Unit	tCO2e
1	Diesel	US-CA	9,200	L	24.66
1	Natural Gas	US-CA	4,300	therm	8.30
1	Biodiesel (biogenic)	US-CA	800	L	2.14
2	Electricity	US-CA	128,000	kWh	38.14
2	Electricity	US-CA	119,500	kWh	35.61
3	Car	US-CA	46,000	mile	12.66
3	Cloud Services	US-CA	88,000	USD	7.04
3	Flight Long Haul	GLOBAL	21,000	mile	6.59
3	Landfill	US-CA	5,400	kg	2.52

3	Truck	US-CA	12,000	tonne-km	1.07
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Rows marked (biogenic) are combustion of biofuels: their CO2 is reported as a separate biogenic memo item and is excluded from the fossil Scope 1 total shown elsewhere in this report, per the GHG Protocol.

12. Data Quality & Completeness

Activity records	10
Complete records	10 of 10
Completeness	100%
Distinct emission factors	8

13. Management Declaration, Assurance & Limitations

Management responsibility statement

Management of Northwind Manufacturing, Inc. is responsible for the preparation of this greenhouse gas inventory, for the completeness and accuracy of the underlying activity data, and for maintaining adequate internal controls over its GHG accounting. This inventory has been prepared in accordance with the GHG Protocol Corporate Accounting and Reporting Standard, applied consistently.

California SB 253 (the Climate Corporate Data Accountability Act) requires covered entities to obtain third-party assurance of their reported emissions - limited assurance for Scope 1 and Scope 2, phasing to reasonable assurance, with Scope 3 assurance following. This report is structured to support that assurance: it provides the organizational boundary, methodology and formulae, a complete emission-factor audit trail, Scope 3 category coverage, and the underlying activity data an assurance provider requires.

This document does not itself constitute an assurance opinion, legal advice, or a regulatory filing. The organization is responsible for the completeness and accuracy of the input data and for engaging a qualified third-party assurance provider prior to submission.

Authorized officer (signature)

Date

Name / title:

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